

Assignment 16: Student Record Management System (Salesforce Apex + Visualforce)

Step 1: Create Custom Object

Go to: **Setup** → **Object Manager** → **Create** → **Custom Object**

Create object:

- **Label:** Student
- **Plural Label:** Students
- **Object Name:** Student
- **Record Name:** Student Name
- **Data Type:** Text

Click **Save**.

Step 2: Create Custom Fields

Inside Student object, create these fields:

1. Roll No

- Data Type: Number
- Field Label: Roll No
- Length: 10
- Decimal Places: 0
- API Name: `Roll_No__c`

2. Class

- Data Type: Text
- API Name: `Class__c`

3. Mobile No

- Data Type: Phone
- API Name: `Mobile_No__c`

Default Salesforce field:

- Student Name → `Name`
-

Step 3: Create Apex Controller Class

Go to: **Setup** → **Apex Classes** → **New**

Class Name:

Paste code:

```
public class StudentController {

    public Student__c student {get; set;}

    public StudentController() {
        student = new Student__c();
    }

    public void saveStudent() {
        insert student;
        student = new Student__c();
    }

    public List<Student__c> getStudents() {
        return [
            SELECT Id, Name, Roll_No__c, Class__c, Mobile_No__c
            FROM Student__c
        ];
    }

    public PageReference deleteStudent() {
        String sid = ApexPages.currentPage().getParameters().get('sid');

        Student__c s = [
            SELECT Id
            FROM Student__c
            WHERE Id = :sid
            LIMIT 1
        ];

        delete s;
        return null;
    }
}
```

Click **Save**.

Step 4: Create Visualforce Page

Go to: **Setup** → **Visualforce Pages** → **New**

Page Name:

Paste code:

```
<apex:page controller="StudentController">
  <apex:form>
    <apex:pageBlock title="Student Record Management">

      <apex:pageBlockSection columns="1">
        <apex:inputField value="{!student.Name}" />
        <apex:inputField value="{!student.Roll_No__c}" />
        <apex:inputField value="{!student.Class__c}" />
        <apex:inputField value="{!student.Mobile_No__c}" />
      </apex:pageBlockSection>

      <apex:commandButton value="Save Student" action="{!
saveStudent}" />

      <apex:pageBlockTable value="{!students}" var="s">
        <apex:column value="{!s.Name}" headerValue="Student Name" />
        <apex:column value="{!s.Roll_No__c}" headerValue="Roll No" />
        <apex:column value="{!s.Class__c}" headerValue="Class" />
        <apex:column value="{!s.Mobile_No__c}" headerValue="Mobile
No" />

        <apex:column>
          <apex:commandLink value="Delete" action="{!
deleteStudent}">
            <apex:param name="sid" value="{!s.Id}" />
          </apex:commandLink>
        </apex:column>
      </apex:pageBlockTable>

    </apex:pageBlock>
  </apex:form>
</apex:page>
```

Click **Save**.

Step 5: Run Application

Open the Visualforce page preview.

Enter:

- Student Name
- Roll No
- Class

- Mobile No

Click: **Save Student**

Records will display in the table.

To remove a student: Click: **Delete**

Features Implemented

- Add Student
 - View Students
 - Delete Student
 - Frontend using Visualforce
 - Backend using Apex
 - Custom object storage
-

Viva Points

- `Student__c` is a custom object.
- `Name` is Salesforce default record name field.
- Apex controller handles backend logic.
- Visualforce provides frontend UI.
- SOQL fetches records.
- DML operations used: insert and delete.